# A brief report on Data analysis of top 50 U.S Universities (2026)

## 1. Introduction

This project analyses a dataset containing information about the top 50 university in the united states for 2026. The dataset includes attributes such as: university name, national rank, founded year, institution type, state, research impact score, international student ratio and employment rate.

The purpose for this analysis is to identify patterns, relationships and hidden structures within the dataset using data preprocessing techniques.

## 2. Methodology

### a) Data preprocessing

Before performing the analysis, the dataset was preprocessed to ensure data quality and consistency.

The following were the steps applied:

- ✓ *Data cleaning*: Duplicate records were removed and checked for missing values. Since our data didn't have any missing values we dived into encoding.
- ✓ *Encoding categorical Variables*: Categorical features such as university name, institution type and state were converted into numerical form using one-hot encoding so that they could be used in machine learning algorithms.
- ✓ *Data normalization*: Numerical attributes such as national rank, , founded year, research impact score and employment rate were scaled using MinMaxScaler to bring all values within a similar range.

These steps ensured that the dataset was structured and suitable for further analysis.

### b) Association Rule Mining

This was applied using the apriori algorithm to discover relationships between different attributes in the dataset. The algorithm identifies frequent itemsets and generates rules that describe how certain features are related to each other using metrics such as:

- ✓ *Support*
- ✓ *Confidence*
- ✓ *Lift*

These metrics help determine the strength and reliability of relationships between variables.

c) Clustering Analysis

This was performed using the K-Means clustering algorithm to identify natural groupings among universities based on their characteristics. The clustering process involved:

- ✓ Selecting important numerical features such as national rank, founded year, research impact score, international student ratio and employment rate.
- ✓ Applying the Elbow Method to determine the optimal number of clusters.
- ✓ Running the K-Means algorithm to group universities into clusters with similar characteristics.

Clustering helps reveal hidden structures and similarities between the institutions.

## 3. Observations

Several important observations were identified from the analysis:

i. Universities with higher research impact scores tend to have better national rankings.
ii. Institutions with higher international student ratios often show stronger global reputation and diversity.
iii. Universities with higher employment rates tend to be associated with strong academic programs and industry connections.
iv. Universities naturally grouped into different categories such as high-performing research institutions, balanced academic institutions and institutions with strong employment outcomes.

The association rules further revealed relationships between ranking, research performance and student outcomes.

## 4. Conclusion

This project successfully applied data mining techniques to analyze the Top 50 U.S. Universities dataset. Through data preprocessing, association rule mining and clustering analysis, meaningful patterns and insights were discovered.

The results demonstrate that research impact, employment rate and international student presence play significant roles in distinguishing top universities. Clustering analysis helped categorize universities based on their overall characteristics while association rules highlighted relationships between key attributes.

Overall, data mining techniques proved useful in uncovering hidden insights within the dataset and providing a deeper understanding of the factors that contribute to university performance and ranking.